(MATH101, 102, 311, 312) Calculus and Analysis

Problems

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Contents

1 (MATH101) Calculus I

Problems within each course are arranged chronologically by year and semester, then by institution in the order POSTECH, KAIST, and SNU, while preserving the original problem order within each source.

Problem 1.1. (2021 Spring KAIST Calculus I) Does each of the following infinite series converge? Mark Yes or No. You do not need to justify your answers.

(1) $\displaystyle\sum_{n=1}^{\infty} \frac{n!}{n^n}$

(2) $\displaystyle\sum_{n=1}^{\infty} \frac{e^{\sqrt{n}}}{n\sqrt{n}}$

(3) $\displaystyle\sum_{n=1}^{\infty} \left(1-\frac{1}{n}\right)^{n}$

(4) $\displaystyle\sum_{n=1}^{\infty} (-1)^n \left(1-\frac{1}{n}\right)^{n\ln n}$

(5) $\displaystyle\sum_{n=1}^{\infty} \left(1-\frac{1}{n}\right)^{2n\ln n}$

(6) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{(n+1)\ln(n+1)}$

(7) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{(n+1)\ln(n+1)\ln\ln(n+1)}$

(8) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{(n+1)(\ln(n+1))^2}$

(9) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{\tan\left(\arccos\left(\frac{1}{n+1}\right)\right)}$

(10) $\displaystyle\sum_{n=1}^{\infty} \frac{\sin(1/n)}{n^{\alpha}}, \qquad \alpha > 0$

Problem 1.2. (2021 Spring KAIST Calculus I) Find the area of the portion of the ellipse

$$9x^2 + 16y^2 = 144$$

between the y-axis and $x = 2$.

Problem 1.3. (2021 Spring KAIST Calculus I) Find the antiderivative of

$$\frac{1}{(x^2+x+1)(x^2+x+2)(x^2+x+3)}.$$

Problem 1.4. (2021 Spring KAIST Calculus I) Does the following improper integral converge?

$$\int_0^{\infty} \frac{\sin^3 x}{x}\, dx.$$

Problem 1.5. (2021 Spring KAIST Calculus I) Prove or disprove:

(1) If $\displaystyle\sum_{n=1}^{\infty} a_n$ converges, then $\displaystyle\sum_{n=1}^{\infty} a_n^2$ also converges.

(2) If both $\displaystyle\sum_{n=1}^{\infty} a_n$ and $\displaystyle\sum_{n=1}^{\infty} |b_n|$ converge, then $\displaystyle\sum_{n=1}^{\infty} a_n b_n$ also converges.

(3) Assume $a_n \neq 0$ for all n. If $\displaystyle\lim_{n\to\infty} a_n = 0$ and $\displaystyle\sum_{n=1}^{\infty} |\sin(a_n)|$ converges, then $\displaystyle\sum_{n=1}^{\infty} |a_n|$ converges.

Problem 1.6. (2021 Spring KAIST Calculus I) Evaluate

$$\int_1^\infty \left(2\sqrt{x^2-1} - 2x + \frac{1}{x}\right) \mathrm{d}x.$$

Problem 1.7. (2021 Spring KAIST Calculus I) Let f be a continuously differentiable (i.e., $f(x)$ is differentiable and $f'(x)$ is continuous) one-to-one function defined on $[a, b]$. Assume that $f'(x) \neq 0$ on $[a, b]$. Compute the following sum of two integrals:

$$\int_{f(a)}^{f(b)} f^{-1}(y)\ \mathrm{d}y + \int_a^b f(x)\ \mathrm{d}x.$$

Problem 1.8. (2021 Spring KAIST Calculus I) Let n be a positive integer and denote the following improper integral by a_n:

$$a_n = \int_0^1 (\ln x)^n\ \mathrm{d}x.$$

(a) Show that the improper integral a_n is convergent. Compute the value of a_n.

(b) Is the series

$$\sum_{n=1}^{\infty} \frac{1}{a_n}$$

absolutely convergent, conditionally convergent, or divergent? Explain the reason using appropriate tests.

(c) Is the series

$$\sum_{n=2}^{\infty} (-1)^n \frac{1}{\ln |a_n|}$$

absolutely convergent, conditionally convergent, or divergent? Explain the reason using appropriate tests.

Problem 1.9. (2021 Spring KAIST Calculus I) Show that the power series

$$\sum_{n=2}^{\infty} n(n-1)x^n, \qquad |x| < 1,$$

can be expressed as

$$\frac{x^2}{(1-x)^3}.$$

Problem 1.10. (2021 Spring KAIST Calculus I) Consider the standard ellipse C,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \qquad a > b > 0.$$

Denote by e the eccentricity of C.

(a) Find an upper bound of the function

$$f(t) = \sqrt{1 - e^2 \sin^2 t}$$

using the first two terms of its binomial series expansion.

(b) Show that the perimeter (total arc length) of C is less than

$$2\pi a \left(1 - \frac{e^2}{4}\right).$$

Problem 1.11. (2021 Spring KAIST Calculus I) Consider the power series

$$\sum_{n=1}^{\infty} (-1)^n \frac{(x-1)^{2n+1}}{n2^n}.$$

(a) Find its interval of convergence.

(b) At each point in the interval of convergence, discuss its mode of convergence as absolute or conditional convergence together with proper reasoning.

Problem 1.12. (2021 Fall KAIST Calculus I) Is each of the following statements true or false? Mark T or F. You do not need to justify your answers.

(1) $f = O(g)$ implies $f = o(g)$ for functions f and g that are positive for x sufficiently large.

(2) $x(\ln x)^{2022} = o(x^{1.001})$.

(3) If $f'(a) \neq 0$, then there exists a $\delta > 0$ such that $f(x) \neq f(a)$ for all $x \neq a$ with $a - \delta < x < a + \delta$.

(4) $\displaystyle\int_2^{\infty} \frac{1 - 2^{-x}}{1 + \sqrt{x}}\, dx$ converges.

(5) $\displaystyle\int_1^{6} \frac{1}{x - 4}\, dx$ converges.

(6) $\displaystyle\int_2^{\infty} \frac{1}{\ln x}\, dx$ converges.

(7) $\displaystyle\int_1^{2} \frac{1}{x \ln x}\, dx$ converges.

(8) $\displaystyle\int_{-\infty}^{\infty} \frac{1}{e^x + e^{-x}}\, dx$ converges.

(9) $\displaystyle\int_{-\infty}^{\infty} \frac{2x}{x^2 + 1}\, dx$ converges.

(10) $\displaystyle\lim_{x \to \infty} \frac{\ln(x + 10^{2022})}{\ln x} = 1.$

Problem 1.13. (2021 Fall KAIST Calculus I) Evaluate

$$\int_1^{\infty} \left(2\sqrt{x^2 - 1} - 2x + \frac{1}{x}\right) dx.$$

Problem 1.14. (2021 Fall KAIST Calculus I) Let f be a continuously differentiable (i.e., $f(x)$ is differentiable and $f'(x)$ is continuous) one-to-one function defined on $[a, b]$. Assume that $f'(x) \neq 0$ on $[a, b]$. Compute the following sum of two integrals:

$$\int_{f(a)}^{f(b)} f^{-1}(y)\ \mathrm{d}y + \int_a^b f(x)\ \mathrm{d}x.$$

Problem 1.15. (2021 Fall KAIST Calculus I)

(a) Derive $\dfrac{\mathrm{d}}{\mathrm{d}x}(\operatorname{arctanh} x)$ for $|x| < 1$.

(b) Find $\displaystyle\int x \operatorname{arctanh} x\ \mathrm{d}x$.

Problem 1.16. (2021 Fall KAIST Calculus I) Let

$$\Gamma(n) = \int_0^\infty x^{n-1} e^{-x}\ \mathrm{d}x$$

for a positive integer n.

(a) Use a comparison test to show that $\Gamma(n)$ converges.

(b) Show that $\Gamma(n) = (n-1)!$.

Problem 1.17. (2021 Fall KAIST Calculus I) Let $w(t)$ be a differentiable function and let κ and ρ be positive constants.

(a) Solve

$$\frac{\mathrm{d}}{\mathrm{d}t} w(t) + \kappa w(t) - \rho = 0 \qquad \text{or} \qquad w' + \kappa w - \rho = 0$$

with the initial condition $w(0) = v_0$.

(b) By setting $w(t) = \dfrac{\mathrm{d}}{\mathrm{d}t} y(t)$, solve

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2} y(t) + \kappa \frac{\mathrm{d}}{\mathrm{d}t} y(t) - \rho = 0 \qquad \text{or} \qquad y'' + \kappa y' - \rho = 0$$

with $y(0) = 0$ and

$$\left.\frac{\mathrm{d}}{\mathrm{d}t} y(t)\right|_{t=0} = v_0.$$

Problem 1.18. (2021 Fall KAIST Calculus I)

(1) Solve the equation

$$xy' + 2y = \frac{8x}{(x-1)(x+1)(x+2)}$$

for $x > 1$.

(2) Solve the initial value problem

$$y' = 2x\sqrt{1 + y^2}, \qquad y(0) = 0.$$

Problem 1.19. (2021 Fall KAIST Calculus I) Determine the radius and interval of convergence of the following series.

(1) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n} x^n$

(2) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n!} x^n$

(3) $\displaystyle\sum_{n=0}^{\infty} (-1)^n x^{2^n}$

Problem 1.20. (2021 Fall KAIST Calculus I) Consider the power series

$$\sum_{n=1}^{\infty} (-1)^n \frac{(x-1)^{2n+1}}{n2^n}.$$

(a) Find its interval of convergence.

(b) At each point in the interval of convergence, discuss its mode of convergence as absolute or conditional convergence together with proper reasoning.

2 (MATH102) Calculus II

Problem 2.1. (2020 Fall KAIST Calculus II) A particle is located at the point $(2, -3)$ on a metal plate whose temperature is given by

$$T(x, y) = 20 - 4x^2 - y^2.$$

Find the directions of

(a) the most rapid increase in temperature,
(b) the most rapid decrease in temperature, and
(c) the least change.

Problem 2.2. (2020 Fall KAIST Calculus II) Evaluate

$$\iint_R (2x - y^2) \, \mathrm{d}A,$$

where R is the region enclosed by the lines

$$y = -x + 1, \qquad y = x + 1, \qquad y = 3.$$

Problem 2.3. (2020 Fall KAIST Calculus II) Consider the function

$$f(x, y) = \begin{cases} \dfrac{x^3 + y^2}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

(a) Determine whether the limit
$$\lim_{(x,y)\to(0,0)} f(x, y)$$
exists. If it exists, evaluate it. If it does not, explain why.
(b) Let $\mathbf{u} = (u_1, u_2)$ be a unit vector in $\mathbb{R}^2$, and let $D_{\mathbf{u}} f(0, 0)$ denote the directional derivative of f along $\mathbf{u}$ at the origin. Determine whether $D_{\mathbf{u}} f(0, 0)$ exists. If it exists, evaluate it. If it does not, explain why. It is possible that $D_{\mathbf{u}} f(0, 0)$ exists for certain $\mathbf{u}$ but does not exist for others.

Problem 2.4. (2020 Fall KAIST Calculus II) The surfaces

$$(x + y)^2 + z - 5 = 0 \qquad \text{and} \qquad (x - y)^2 - z - 5 = 0$$

intersect along a closed curve C.

(a) Find the point on C at which the function $2x + y$ is maximized.
(b) Find a parametric equation of the tangent line to C at the point $(1, -2, 4)$.

Problem 2.5. (2020 Fall KAIST Calculus II) We are given

$$w = e^{xy+z^2}, \qquad x = r + s, \qquad y = r^2 s, \qquad z = \sin(r + s).$$

Find

$$\frac{\partial w}{\partial r}$$

when $(r, s) = (1, 1)$.

Problem 2.6. (2020 Fall KAIST Calculus II) Evaluate

$$\int_0^3 \int_{\sqrt{x/3}}^1 e^{y^3}\, dy\, dx.$$

Problem 2.7. (2020 Fall KAIST Calculus II) Find $\mathbf{T}$, $\mathbf{N}$, and the curvature κ for the curve

$$\mathbf{r}(t) = (5\cos 3t, 6t, 5\sin 3t).$$

Problem 2.8. (2020 Fall KAIST Calculus II) Let R be the set of points (x, y) in the xy-plane with $-2 < x < 2$. For each $(x, y) \in R$ with $(x, y) \neq (0, 0)$, let

$$f(x, y) = \frac{(y + \sin x)y^2}{y^4 + (y + \sin x)^2},$$

and let $f(0, 0) = 0$.

(a) Find $\nabla f(0, 0)$.

(b) Prove or disprove that f is continuous at $(0, 0)$.

Problem 2.9. (2020 Fall KAIST Calculus II) Let H be the hyperboloid in $\mathbb{R}^3$ determined by

$$2y^2 - x^2 - z^2 = 3,$$

and let E be the ellipsoid in $\mathbb{R}^3$ determined by

$$x^2 + 2(y - 1)^2 + z^2 = 7.$$

(a) Find an equation for the tangent plane to H at the point $P(1, 2, -2)$.

(b) Identify the curve C which is the intersection of H and E. Find the line tangent to C at P.

Problem 2.10. (2020 Fall KAIST Calculus II) For each (x, y) in the xy-plane, let

$$f(x, y) = e^{x+2y}\cos(xy).$$

Find the second Taylor polynomial of f about the point $(1, \pi)$.

Problem 2.11. (2020 Fall KAIST Calculus II) A rectangular box is to be inscribed in the sphere

$$x^2 + y^2 + z^2 = 4.$$

Using the Lagrange multiplier method, find the volume of a box having maximum surface area.

Problem 2.12. (2020 Fall KAIST Calculus II) Find the smallest and largest values of

$$f(x, y) = (x + 2y)e^{x+y}$$

on the region

$$\{(x, y) : -1 \le x \le 1,\ -1 \le y \le 1\}.$$

Problem 2.13. (2020 Fall KAIST Calculus II) Find all critical points of

$$f(x, y) = e^x(x^2 - y^2)$$

and classify them.

Problem 2.14. (2020 Fall KAIST Calculus II) Use a double or surface integral to answer the following.

(a) Let S be the surface described by

$$z = x^2 + y^2, \qquad 0 \le z \le 1.$$

Find the area of S using the parametrization

$$\mathbf{r}(x, y) = (x, y, x^2 + y^2).$$

(b) Consider the surface given by the parametrization

$$\mathbf{r}(u, v) = \left(\frac{u^2}{2} + v, uv, v^2\right), \qquad 0 \le u, v \le 1.$$

Write the integral which expresses the surface area. You do not have to evaluate it.

Problem 2.15. (2020 Fall KAIST Calculus II) Let C be the curve parametrized by

$$\mathbf{r}(t) = (\cos t, \sin t, e^{\cos t}), \qquad t \in [0, \pi].$$

Let $\mathbf{F}(x, y, z)$ be a vector field in $\mathbb{R}^3$ satisfying

$$\int_C \mathbf{F} \cdot \mathrm{d}\mathbf{r} = \int_C x\ \mathrm{d}y - x\ \mathrm{d}z + z^2\ \mathrm{d}x.$$

Evaluate

$$\int_C \mathbf{F} \cdot \mathrm{d}\mathbf{r}.$$

Problem 2.16. (2020 Fall KAIST Calculus II) Find the point on the ellipsoid

$$3x^2 + 2y^2 + z^2 = 6$$

which is farthest from the plane

$$x + y - z = 1.$$

Problem 2.17. (2020 Fall KAIST Calculus II) Compute

$$\int_{-3}^{3} \int_{x^2}^{9} \int_{-\sqrt{y-x^2}}^{\sqrt{y-x^2}} \sqrt{x^2 + z^2}\ \mathrm{d}z\ \mathrm{d}y\ \mathrm{d}x.$$

Problem 2.18. (2020 Fall KAIST Calculus II) Let X be the set of points $(x, y, z) \in \mathbb{R}^3$ satisfying

$$r = r(x, y, z) = \sqrt{x^2 + y^2 + z^2} > 0.$$

Let m be a constant. Define a vector field $\mathbf{F}$ on X by

$$\mathbf{F}(x, y, z) = \left(\frac{x}{r^m}, \frac{y}{r^m}, \frac{z}{r^m}\right).$$

(a) Find the values of m, if any, for which $\mathbf{F}$ is irrotational, i.e.,

$$\nabla \times \mathbf{F} = \mathbf{0}$$

on X.

(b) Find the values of m, if any, for which $\mathbf{F}$ is incompressible, i.e.,

$$\nabla \cdot \mathbf{F} = 0$$

on X.

(c) For each m, prove or disprove that $\mathbf{F}$ is the curl of a vector field whose components have continuous partial derivatives.

Problem 2.19. (2020 Fall KAIST Calculus II) Find the centroid of the solid enclosed by the hemisphere

$$x^2 + y^2 + (z-1)^2 = 1, \qquad z \geq 1,$$

and the cone

$$z = \sqrt{x^2 + y^2}.$$

Problem 2.20. (2020 Fall KAIST Calculus II) Let S be the portion of the sphere

$$x^2 + y^2 + (z-1)^2 = 2, \qquad z \geq 0.$$

Let

$$\mathbf{F}(x, y, z) = (y, -x, e^{xz}).$$

Evaluate

$$\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, \mathrm{d}S,$$

where

$$\mathbf{n}(0, 0, 1 + \sqrt{2}) = (0, 0, 1).$$

Problem 2.21. (2020 Fall KAIST Calculus II) Let

$$S_1 = \{(x, y, z) : x^2 + y^2 = 1\}$$

be the cylinder and

$$S_2 = \{(x, y, z) : z = 2xy\}$$

be the hyperbolic paraboloid, and let C be the intersection curve of S_1 and S_2. Let $\mathbf{r}(t)$ be a parametrization of the portion of C with initial point $(1, 0, 0)$ and endpoint

$$\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 1\right).$$

Denote this portion of C by C_1. Let $\mathbf{F}$ be the vector field

$$\mathbf{F}(x, y, z) = \left(\sec^2(x+y) + yz^2 + e^{y+z}, \sec^2(x+y) + xe^{y+z} + xz^2, xe^{y+z} + xy\right).$$

Evaluate

$$\int_{C_1} \mathbf{F} \cdot \mathrm{d}\mathbf{r}.$$

Problem 2.22. (2020 Fall KAIST Calculus II) Let S be the part of the ellipsoid

$$S = \left\{ (x, y, z) : \frac{x^2}{4} + \frac{y^2}{4} + \frac{z^2}{2} = 1,\ z \geq 1 \right\},$$

oriented with outward normal vectors (looking from the origin). Compute the flux of the vector field $\mathbf{F}$ across S, where

$$\mathbf{F}(x, y, z) = \left(ze^{yz}, y\sin(x^3) - 3x^2, 2z^2 - z\sin(x^3) + ye^{(x^2+y^2)^2} \right).$$

Problem 2.23. (2020 Fall KAIST Calculus II) Find the mass of a spring having the shape of a portion C of the circular helix

$$\mathbf{r}(t) = \frac{1}{\sqrt{2}}(\cos t, \sin t, t), \qquad 0 \leq t \leq 6\pi,$$

and density

$$\delta(x, y, z) = 1 + z.$$

Problem 2.24. (2020 Fall KAIST Calculus II) Use Green's theorem to find the work done by the force

$$\mathbf{F}(x, y) = (y^3, x^3 + 3xy^2)$$

on a particle while it travels once around the circle of radius 3 centered at the origin.

Problem 2.25. (2020 Fall KAIST Calculus II) A cone-shaped surface lamina S is given by

$$z = 4 - 2\sqrt{x^2 + y^2}, \qquad 0 \leq z \leq 4.$$

At each point of S, the density is equal to the distance between the point and the z-axis. Find the mass of the lamina.

Problem 2.26. (2020 Fall KAIST Calculus II) Find the volume of material cut from the solid sphere

$$r^2 + z^2 \leq 9$$

by the cylinder

$$r = 3\sin\theta$$

in cylindrical coordinates in $\mathbb{R}^3$.

Problem 2.27. (2020 Fall KAIST Calculus II) Find the outward flux of

$$\mathbf{F}(x, y, z) = (x, y, z)$$

across the boundary of the region

$$D = \left\{ (x, y, z) : \sqrt{x^2 + y^2} \leq z \leq \sqrt{2 - x^2 - y^2} \right\}.$$

Problem 2.28. (2021 Spring KAIST Calculus I) Mark True or False for each of the following statements. You do not need to justify your answers.

(1) If a series $\sum_{n=0}^{\infty} a_n x^n$ converges on $[-1,1]$, then the series $\sum_{n=1}^{\infty} n a_n x^{n-1}$ converges on $(-1,1)$.

(2) If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are three-dimensional vectors, then

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) \times \mathbf{c}.$$

(3) If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are nonzero three-dimensional vectors and $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = 0$, then $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are in the same plane.

(4) Assume f is infinitely differentiable at 0 (i.e., $f^{(n)}(0)$ exists for all $n = 1, 2, 3, \ldots$). Then there exists $\varepsilon > 0$ such that

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

for all $-\varepsilon < x < \varepsilon$.

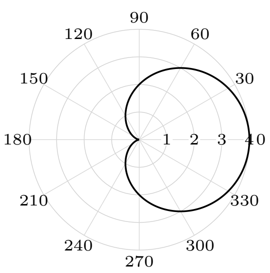
(5) Let $A(x) = \sum_n a_n x^n$ and $B(x) = \sum_n b_n x^n$ be power series with radii of convergence R_1 and R_2, respectively. Then $A(x)B(x)$ is again a power series with radius of convergence $\min\{R_1, R_2\}$.

Problem 2.29. (2021 Spring KAIST Calculus I) Match each of the following polar equations with an appropriate graph.

(1) $r = 4\cos(2\theta)$

(2) $r = \dfrac{4}{2 - \cos\theta}$

(3) $r = 2(1 + \cos\theta)$



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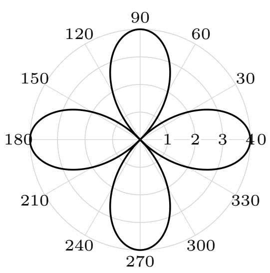

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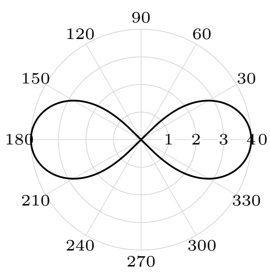

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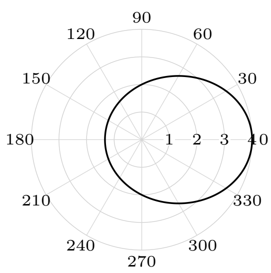


(4)

Problem 2.30. (2021 Spring KAIST Calculus I) Let

$$L_1 : \ x = 1 + t, \ y = 3 - t, \ z = 2t, \qquad L_2 : \ x = 0, \ y = -2s, \ z = 5s.$$

Find the minimum of $\|\overrightarrow{P_1P_2}\|$ for $P_1 \in L_1$ and $P_2 \in L_2$.

Problem 2.31. (2021 Spring KAIST Calculus I) Let M_1 be the plane in space given by

$$2x - y - 3z = 2,$$

and let M_2 be the plane perpendicular to M_1 through the points $P(1, -2, -4)$ and $Q(0, -1, -2)$.

(a) Find parametric equations for the line L of intersection of M_1 and M_2.

(b) Find an equation for a plane M parallel to L that contains the line through the points $A(-2, -4, 3)$ and $B(2, 0, 4)$. Find the distance between L and M.

Problem 2.32. (2021 Spring KAIST Calculus I) For the surface with equation

$$(y - \sqrt{2z_0})^2 - x^2 = z,$$

consider its curve cut C by the plane $z = z_0 > 0$.

(a) Identify the type of C and find its eccentricity.

(b) Find the foci and directrices of the curve cut on the plane $z = z_0$.

(c) Find the polar equation $r = f(\theta)$ of the curve cut on the plane $z = z_0$.

Problem 2.33. (2021 Fall POSTECH Calculus II) Find an equation of the plane that passes through the line of intersection of the planes

$$x + y = 2 \qquad \text{and} \qquad 2x - y + 3z = 1$$

and passes through the point $(1, 2, 1)$.

Problem 2.34. (2021 Fall POSTECH Calculus II) Let $f(x, y) = |xy|$. Show that f is differentiable at $(0, 0)$.

Problem 2.35. (2021 Fall POSTECH Calculus II) Let $F(x, y, z)$ be differentiable everywhere. Assume that

$$F(1 + t^2, 2t, -1 - t) = 0$$

for all t, and that $\nabla F(2, 2, -3) = (a, b, c)$. Show that

$$a + b = \frac{c}{2}.$$

Problem 2.36. (2021 Fall POSTECH Calculus II) A plane P contains the points $(-3, 0, 0)$, $(0, -3, 0)$, and $(0, 0, -3)$. Find all points on

$$z = x^2 + y^2$$

whose tangent plane does not intersect P.

Problem 2.37. (2021 Fall POSTECH Calculus II) Find the absolute maximum and minimum values of

$$f(x, y) = x^4 + y^4 - 4xy + 1$$

on the region

$$x^4 + y^4 \leq 1.$$

Problem 2.38. (2021 Fall POSTECH Calculus II) Let L_1 and L_2 be lines in $\mathbb{R}^3$ parametrized by

$$L_1(t) = (1 + t, 1 - t, 2t), \qquad L_2(s) = (-s, s, -s),$$

respectively. Find the shortest distance between L_1 and L_2.

Problem 2.39. (2021 Fall POSTECH Calculus II) Let $f(x, y) = |xy|$. Show that there is no open ball centered at $(0, 0)$ in which f_x and f_y are defined and continuous.

Problem 2.40. (2021 Fall POSTECH Calculus II) Let

$$f(x,y) = \begin{cases} \dfrac{(xy)^\alpha}{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

For which $\alpha > 0$ is f differentiable in $\mathbb{R}^2$?

Problem 2.41. (2021 Fall POSTECH Calculus II) Let

$$R = \left\{ (x,y) : 0 \leq x \leq \frac{\sqrt{3}}{2},\ 0 \leq y \leq \frac{\sqrt{3}}{2},\ x^2+y^2 \leq 1 \right\}.$$

Find the minimum and maximum of the linear function

$$F(x,y) = -x + 2y.$$

Justify your answer.

Problem 2.42. (2021 Fall POSTECH Calculus II) Show that

$$f(x,y) = x^2 - 2xy^2 - y^4$$

has a saddle point at $(0,0)$.

Problem 2.43. (2021 Fall POSTECH Calculus II) Show by giving a counterexample that the cross product is not associative; namely, find three vectors $\mathbf{U}, \mathbf{V}, \mathbf{W} \in \mathbb{R}^3$ such that

$$(\mathbf{U} \times \mathbf{V}) \times \mathbf{W} \neq \mathbf{U} \times (\mathbf{V} \times \mathbf{W}).$$

Problem 2.44. (2021 Fall POSTECH Calculus II) Determine the maximum and minimum values of

$$f(x,y) = xy \ln(1 + x^2 + y^2)$$

in the region

$$x^2 + y^2 \leq 1.$$

Problem 2.45. (2021 Fall POSTECH Calculus II) Evaluate

$$I_x = \iiint_D x \, \mathrm{d}V,$$

where

$$D = \left\{ (x,y,z) : 1 \leq x^2+y^2+z^2 \leq 4,\ x \geq 0 \right\}.$$

Problem 2.46. (2021 Fall POSTECH Calculus II) Evaluate the surface integral

$$\iint_S \mathbf{F} \cdot \mathbf{N} \, \mathrm{d}S$$

representing the flux of the vector field

$$\mathbf{F}(x,y,z) = (1+x, x+2y, -3x+z)$$

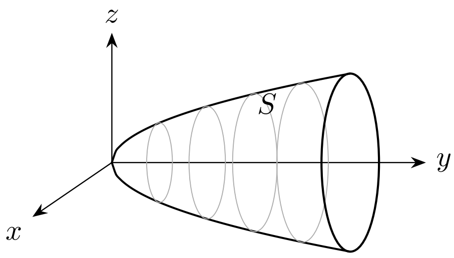
across the hemisphere

$$S = \left\{ (x,y,z) : x^2+y^2+z^2 = 1,\ x \geq 0 \right\}$$

in the direction of the unit normal vector field whose first component is positive.

Problem 2.47. (2021 Fall POSTECH Calculus II) Let $S \subseteq \mathbb{R}^3$ be the surface defined by

$$y = x^2 + z^2, \qquad x^2 + z^2 \leq 1.$$



Let

$$\mathbf{F}(x, y, z) = (y, z, x).$$

Find the inward flux of $\nabla \times \mathbf{F}$ across S in the direction of the unit normal vector field whose second component is positive.

Problem 2.48. (2021 Fall POSTECH Calculus II) Let

$$f(x, y) = |(x-1)(y-1)|.$$

Show that there is no open ball centered at $(1, 1)$ in which f_x and f_y are defined and continuous.

Problem 2.49. (2021 Fall POSTECH Calculus II) Find the volume of the solid E bounded by both the sphere

$$x^2 + y^2 + z^2 = 2z$$

and the sphere

$$x^2 + y^2 + z^2 = \frac{1}{4}.$$

Problem 2.50. (2021 Fall POSTECH Calculus II) An object moves from the origin to a point that lies on the surface

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

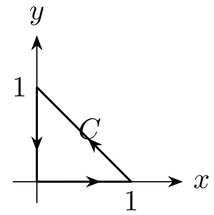
along a smooth curve C under the force field

$$\mathbf{F}(x, y, z) = (yz, zx, xy).$$

Find the maximum possible value of the work integral

$$\int_C \mathbf{F} \cdot \mathrm{d}\mathbf{r}.$$

Problem 2.51. (2021 Fall POSTECH Calculus II) Let C be the triangle shown below, oriented counterclockwise.



Evaluate

$$\oint_C x^2 \, dy.$$

Problem 2.52. (2021 Fall POSTECH Calculus II) Let $S \subseteq \mathbb{R}^3$ be a surface with a parametrization $\boldsymbol{\alpha}\colon \mathcal{R} \to S$ defined by

$$\boldsymbol{\alpha}(u,v) = uv\mathbf{i} + \frac{u}{v}\mathbf{j} + \frac{v}{u}\mathbf{k}.$$

Find an equation for the tangent plane to S at $\boldsymbol{\alpha}(1,1)$.

Problem 2.53. (2021 Fall KAIST Calculus I) Mark True or False for each of the following statements. You do not need to justify your answers.

(1) If a series $\sum_{n=0}^{\infty} a_n x^n$ converges on $[-1,1]$, then the series $\sum_{n=1}^{\infty} n a_n x^{n-1}$ converges on $(-1,1)$.

(2) If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are three-dimensional vectors, then

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) \times \mathbf{c}.$$

(3) If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are nonzero three-dimensional vectors and $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = 0$, then $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are in the same plane.

(4) Assume f is infinitely differentiable at 0 (i.e., $f^{(n)}(0)$ exists for all $n = 1, 2, 3, \ldots$). Then there exists $\varepsilon > 0$ such that

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

for all $-\varepsilon < x < \varepsilon$.

(5) Let $A(x) = \sum_n a_n x^n$ and $B(x) = \sum_n b_n x^n$ be power series with radii of convergence R_1 and R_2, respectively. Then $A(x)B(x)$ is again a power series with radius of convergence $\min\{R_1, R_2\}$.

Problem 2.54. (2021 Fall KAIST Calculus I) Let $\{\mathbf{x}_n\}$ be a sequence of vectors, where

$$\mathbf{x}_n = (x_n^{(1)}, x_n^{(2)}, x_n^{(3)}).$$

Let $\mathbf{x}$ be a vector. Prove or disprove:

(a) $\|\mathbf{x}_n - \mathbf{x}\| \to 0$ if $\sum_{k=1}^{n} \|\mathbf{x}_k - \mathbf{x}\| \leq 1$ for all n.

(b) $\|\mathbf{x}_n - \mathbf{x}\| \to 0$ if

$$\lim_{n\to\infty} \frac{\|\mathbf{x}_{n+1} - \mathbf{x}\|}{\|\mathbf{x}_n - \mathbf{x}\|} = \rho < 1.$$

Problem 2.55. (2021 Fall KAIST Calculus I) How many domains are separately enclosed by the following polar equations?

(1) $r = 4\cos(2\theta)$

(2) $r = \dfrac{4}{2 - \cos\theta}$

(3) $r = 2(1 + \cos\theta)$

Problem 2.56. (2021 Fall KAIST Calculus I) Consider the curve

$$r = \cos 3\theta, \qquad -\frac{\pi}{6} \le \theta \le \frac{\pi}{6}.$$

(a) Find the slopes of the curve at the origin.

(b) Find the length of the curve, using the fact that

$$\int_0^{\pi/3} \sqrt{1 + 8\sin^2 3\theta}\, \mathrm{d}\theta \approx 2.23.$$

(c) Find the area enclosed by the curve.

Problem 2.57. (2021 Fall KAIST Calculus I) Your eye is at $(4, 0, 0)$. You are looking at a triangular plate whose vertices are at $(1, 0, 1)$, $(1, 1, 0)$, and $(-2, 2, 2)$. The line segment from $(1, 0, 0)$ to $(0, 2, 2)$ passes through the plate. What portion of the line segment is hidden from your view by the plate?

Problem 2.58. (2021 Fall KAIST Calculus I) Let

$$L_1: \; x = 1 + t, \; y = 3 - t, \; z = 2t, \qquad L_2: \; x = 0, \; y = -2s, \; z = 5s.$$

Find the minimum of $\|\overrightarrow{P_1P_2}\|$ for $P_1 \in L_1$ and $P_2 \in L_2$.

Problem 2.59. (2022 Spring KAIST Calculus II) Answer the following True/False questions. You do not need to justify your answers.

(a) Let $\mathbf{r}(t)$ be a parametrization of the movement of a particle along a smooth curve in space. If its speed $|\mathbf{r}'(t)|$ is constant, then its principal unit normal vector $\mathbf{N}$ and its acceleration $\mathbf{r}''(t)$ are parallel.

(b) Let $f(x, y)$ be a function of two variables. If the directional derivative $D_{\mathbf{u}}f$ exists for each direction $\mathbf{u}$, then f is continuous at $(0, 0)$.

(c) A smooth space curve $\mathbf{r}(t)$ with zero torsion is necessarily contained in a plane.

(d) Let $\mathbf{r}(t)$ be a smooth curve on the sphere

$$x^2 + y^2 + z^2 = R^2, \qquad R > 0.$$

Then its curvature is always at least $1/R$, i.e., $\kappa \ge 1/R$.

(e) Consider the surface $z = z(x, y)$ given by

$$x^2 + \frac{(y - z)^2}{z^2} = 1, \qquad z > 0.$$

The level curve has an increasing curvature at the origin $(0, 0)$ as the level increases.

Problem 2.60. (2022 Spring KAIST Calculus II) Let C be a smooth curve in space parametrized by $\mathbf{r}(t)$.

(a) Express $\mathbf{r}'$ and $\mathbf{r}''$ in the TNB frame.

(b) Using (a), deduce the following expression for the curvature:

$$\kappa(t) = \frac{|x'y'' - x''y'|}{\left((x')^2 + (y')^2\right)^{3/2}}$$

for $\mathbf{r}(t) = (x(t), y(t))$.

Problem 2.61. (2022 Spring KAIST Calculus II) Consider the ellipse

$$\frac{x^2}{9} + \frac{y^2}{4} = 1.$$

(a) Find the point(s) where the curvature is minimum.

(b) Find the equation of the circle of curvature at the point(s) where the curvature is minimum.

Problem 2.62. (2022 Spring KAIST Calculus II) Let

$$f(x,y) = \begin{cases} \dfrac{4x^2y^2}{x^4 - 2xy^2 + y^3}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

(a) Calculate $f_x(0,0)$ and $f_y(0,0)$.

(b) Calculate the directional derivative $D_{\mathbf{u}}f(0,0)$, where $\mathbf{u}$ is the unit vector in the direction of $(1,1)$.

Problem 2.63. (2022 Spring KAIST Calculus II) Let C be the circular helix in $\mathbb{R}^3$ given by

$$\mathbf{r}(t) = (3\cos t, 3\sin t, 4t)$$

from the point $P_0 = (3,0,0)$ to the point $P_1 = (3,0,8\pi)$.

(a) Find the arc length of C from P_0 to P_1, and reparametrize C by the arc length function $s(t)$.

(b) Find the unit tangent vector $\mathbf{T}$ of C at the point $(-3,0,4\pi)$.

(c) Calculate the curvature κ of C, and find the principal unit normal vector $\mathbf{N}$ of C at the point $(-3,0,4\pi)$.

Problem 2.64. (2022 Spring KAIST Calculus II) Let

$$f(x,y) = \begin{cases} \dfrac{\sin(x^3(y^3+\pi))}{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

Hint. $\displaystyle\lim_{t\to 0} \frac{\sin t}{t} = 1$.

(a) Show that

$$\lim_{(x,y)\to(0,0)} \frac{x^3(y^3+\pi)}{x^2+y^2} = 0.$$

(b) Prove that f is continuous at $(0,0)$.

(c) Calculate $f_x(0,0)$ and $f_y(0,0)$.

Problem 2.65. (2022 Spring KAIST Calculus II) Consider the function

$$f(x, y) = 4x^2y + x^2y^2 + 8xy^2 + 2xy^3 + 1.$$

Find all critical points and classify them.

Problem 2.66. (2022 Spring KAIST Calculus II) Consider the surface C given by

$$z = x^2 + y^2$$

and the point $P = (3, 3, 1)$ in space. Find the equation of the sphere with center P which is tangent to C.

Problem 2.67. (2022 Spring KAIST Calculus II) Find the extreme values of the function

$$f(x, y, z) = xy + z^2$$

on the circle in which the plane $x = y$ intersects the sphere

$$x^2 + y^2 + z^2 = 4.$$

Problem 2.68. (2022 Spring KAIST Calculus II) Consider the function

$$f(x, y) = \cos x \cos y.$$

(a) Use Taylor's formula to find a quadratic approximation near the origin $(0, 0)$.

(b) Estimate the error in the approximation if $|x| \leq 0.1$ and $|y| \leq 0.1$. You do not need to compute exact values; simply find a numerical expression.

Problem 2.69. (2022 Spring KAIST Calculus II) Consider the function $f(x, y)$ given by

$$f(x, y) = \begin{cases} x^m y^n, & \text{if either } x \text{ is rational and } y \text{ is irrational,} \\ & \text{or } x \text{ is irrational and } y \text{ is rational,} \\ 0, & \text{otherwise,} \end{cases}$$

for some nonnegative integers m and n. Let $0^0 = 1$.

(a) Find all nonnegative integers m and n such that f is continuous at $(0, 0)$.

(b) Find all nonnegative integers m and n such that f is differentiable at $(0, 0)$.

Problem 2.70. (2022 Spring KAIST Calculus II) Answer the following True/False questions. You do not need to justify your answers.

(a) Let $\mathbf{F} = (M, N, P)$ be a vector field on $\mathbb{R}^3$ whose components M, N, and P have continuous second partial derivatives. Then

$$\nabla \cdot (\nabla \times \mathbf{F}) = 0.$$

(b) For a function $f\colon [0, 1] \times [0, 1] \to \mathbb{R}$, suppose that the iterated integrals

$$\int_0^1 \int_0^1 f(x, y)\ \mathrm{d}x\ \mathrm{d}y \qquad \text{and} \qquad \int_0^1 \int_0^1 f(x, y)\ \mathrm{d}y\ \mathrm{d}x$$

exist and are equal. Then f is integrable.

(c) Let $\mathbf{F} = (M, N, P)$ be a vector field on an open connected domain $D \subseteq \mathbb{R}^3$ whose components M, N, and P have continuous first partial derivatives and satisfy

$$\frac{\partial N}{\partial x} = \frac{\partial M}{\partial y}, \qquad \frac{\partial P}{\partial y} = \frac{\partial N}{\partial z}, \qquad \frac{\partial M}{\partial z} = \frac{\partial P}{\partial x}.$$

Then $\mathbf{F}$ is conservative on D.

(d) There is a piecewise smooth, simple loop in $\mathbb{R}^2$ whose principal unit normal vector $\mathbf{N}$ and outward unit normal vector $\mathbf{n}$ are coincident.

(e) Let C be a clockwise-oriented, piecewise smooth, simple loop in $\mathbb{R}^2$ whose curvature never vanishes. Let $\mathbf{N}$ be the principal unit normal vector of C. Then, for any vector field $\mathbf{F}$ on $\mathbb{R}^2$ whose components have continuous first partial derivatives,

$$\int_C \mathbf{F} \cdot \mathbf{N} \, \mathrm{d}s + \iint_D \nabla \cdot \mathbf{F} \, \mathrm{d}A = 0,$$

where D is the region surrounded by C.

Problem 2.71. (2022 Spring KAIST Calculus II) Evaluate the integral

$$\int_0^1 \int_x^1 \int_{\sqrt{y}}^1 \frac{e^{z^3}}{z^2} \, \mathrm{d}z \, \mathrm{d}y \, \mathrm{d}x.$$

Problem 2.72. (2022 Spring KAIST Calculus II) Consider the vector field

$$\mathbf{F}(x, y) = \left(-\frac{y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right)$$

on $\mathbb{R}^2 \setminus \{(0, 0)\}$. Calculate the line integral

$$\oint_C \mathbf{F} \cdot \mathrm{d}\mathbf{r}$$

over each of the following curves, oriented counterclockwise.

(a) $C_1 \colon (x - 2)^2 + (y + 1)^2 = 4$.

(b) $C_2 \colon x^2 + y^2 = 9$.

(c) $C_3 \colon r = 9 + 4 \sin 3\theta$, where $(x, y) = (r \cos \theta, r \sin \theta)$.

Problem 2.73. (2022 Spring KAIST Calculus II)

(a) Calculate the divergence of the vector field

$$\mathbf{F}(x, y) = \left(xye^{x^2+y^2}, xye^{x^2+y^2} \right)$$

on the plane $\mathbb{R}^2$.

(b) Calculate the double integral

$$\iint_R (x + y)(1 + 2xy)e^{x^2+y^2} \, \mathrm{d}A,$$

where

$$R = [0, 1] \times [0, 1] = \{(x, y) \in \mathbb{R}^2 : 0 \leq x, y \leq 1\}.$$

Problem 2.74. (2022 Spring KAIST Calculus II) Consider the parallelogram R given by the lines

$$L_1\colon y = \frac{x}{2}, \qquad L_2\colon y = -\frac{5x}{3}, \qquad L_3\colon y = \frac{x+13}{2}, \qquad L_4\colon y = \frac{-5x+13}{3}.$$

Calculate the area $A(R)$ of the parallelogram R in the following steps.

(a) Find a linear transformation $\mathbf{\Phi}\colon \mathbb{R}^2 \to \mathbb{R}^2$ sending the unit square $[0,1]\times[0,1]$ to the parallelogram R.
Hint. If $\mathbf{\Phi}$ is linear, then
$$\mathbf{\Phi}(u,v) = \mathbf{\Phi}(1,0)u + \mathbf{\Phi}(0,1)v$$
for all $(u,v)\in\mathbb{R}^2$.

(b) Calculate $A(R)$ using the transformation $\mathbf{\Phi}$.

Problem 2.75. (2022 Spring KAIST Calculus II) Use cylindrical coordinates to find the volume of the solid in $\mathbb{R}^3$ enclosed by the cone
$$z = \sqrt{x^2+y^2},$$
the cylinder
$$x^2 + (y-1)^2 = 1,$$
and the xy-plane.
Hint. $\sin^3\theta = (1-\cos^2\theta)\sin\theta$.

Problem 2.76. (2022 Spring KAIST Calculus II) Let $\mathbf{h}(x,y)$ be a smooth parametrization on a region H for a surface S in $\mathbb{R}^3$. Suppose that there is a transformation
$$\mathbf{F}\colon R \to H, \qquad (u,v)\mapsto (x(u,v), y(u,v)),$$
such that $\mathbf{F}$ is one-to-one on the interior of the region R, its components have continuous first partial derivatives, and $\mathbf{r} = \mathbf{h}\circ\mathbf{F}$ is a smooth parametrization on R for S. Show that
$$\iint_H |\mathbf{h}_x \times \mathbf{h}_y|\ \mathrm{d}x\ \mathrm{d}y = \iint_R |\mathbf{r}_u \times \mathbf{r}_v|\ \mathrm{d}u\ \mathrm{d}v = A(S). \tag{$\star$}$$
where $A(S)$ is the area of S, in the following steps.

(a) Show that
$$\mathbf{r}_u \times \mathbf{r}_v = (\mathbf{h}_x \times \mathbf{h}_y)\frac{\partial(x,y)}{\partial(u,v)},$$
where $\partial(x,y)/\partial(u,v)$ is the Jacobian of the transformation $\mathbf{F}$.

(b) Use (a) to deduce ($\star$).

Problem 2.77. (2022 Spring KAIST Calculus II) Let D be the solid region with ellipsoidal boundary ∂D:
$$x^2 + y^2 + \frac{z^2}{4} = 9,$$
which is oriented with the unit normal vector pointing outward.

(a) Calculate the volume of D.

(b) Let $\mathbf{F}$ be the vector field on $\mathbb{R}^3$ given by
$$\mathbf{F}(x,y,z) = \left(x^2\sqrt{y^2+z^2}, -y^2\sqrt{x^2+z^2}, z^2\sqrt{x^2+y^2}\right).$$

Calculate the surface integral

$$\iint_{\partial D} \mathbf{F} \cdot \mathbf{n} \, \mathrm{d}S,$$

where $\mathbf{n}$ is the outward unit normal vector.

Problem 2.78. (2022 Spring KAIST Calculus II) Let S be the surface given by

$$x^2 + y^2 + (z-3)^2 = 5^2, \qquad z \geq 0,$$

and let $\mathbf{n}$ be the unit normal vector on S pointing outward from the origin. Calculate the surface integral

$$\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, \mathrm{d}S$$

for the vector field

$$\mathbf{F}(x, y, z) = \left(\frac{x}{1+x^4} - \sin(e^{-y}), \frac{y}{1+y^6} + \sin(e^x), e^{-x^2-y^2-z^2} \right).$$

Hint. For R: $x^2 + y^2 \leq 4$,

$$\iint_R e^x \cos(e^x) \, \mathrm{d}A = \iint_R e^{-y} \cos(e^{-y}) \, \mathrm{d}A.$$

Problem 2.79. (2022 Spring KAIST Calculus II) Consider the function

$$f \colon [0,1] \times [0,1] \subseteq \mathbb{R}^2 \to \mathbb{R}$$

defined by

$$f(x, y) = \begin{cases} \sin(2\pi(x+y)), & \text{if } y \text{ is rational,} \\ -\sin(2\pi(x+y)), & \text{if } y \text{ is irrational.} \end{cases}$$

(a) Show that the iterated integral

$$\int_0^1 \int_0^1 f(x, y) \, \mathrm{d}x \, \mathrm{d}y$$

exists, and evaluate it.

(b) Show that f is not integrable over $[0,1] \times [0,1]$.

Problem 2.80. (2022 Fall POSTECH Calculus II) Let ℓ be a line in $\mathbb{R}^4$ passing through $(0, 1, -1, 3)$ and parallel to $(1, 0, 3, -1)$. If $P = (6, 5, 4, 2)$ is a point not on ℓ, find the distance between ℓ and P.

Problem 2.81. (2022 Fall POSTECH Calculus II) Let $f(x, y) = x^3 + 3y$.

(a) Find an equation of the plane tangent to the graph of f at $(2, 4)$.

(b) Find the linear approximation of $f(x, y)$ near $(2, 4)$, and use it to estimate $f(2.01, 4.03)$.

Problem 2.82. (2022 Fall POSTECH Calculus II) The Laplace operator Δ is defined by

$$\Delta = \partial_{xx} + \partial_{yy}.$$

Compute

$$\Delta \ln\left(\sqrt{x^2 + y^2}\right), \qquad (x, y) \neq (0, 0).$$

Problem 2.83. (2022 Fall POSTECH Calculus II)

(a) Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be linear. Show that there exist numbers C_j, $j = 1, \dots, n$, such that

$$f(\mathbf{x}) = \sum_{j=1}^{n} C_j x_j, \qquad \mathbf{x} = (x_1, \dots, x_n).$$

(b) Show that

$$f(\mathbf{x}) = \sum_{j=1}^{n} C_j x_j$$

is differentiable on $\mathbb{R}^n$. Find ∇f.

Problem 2.84. (2022 Fall POSTECH Calculus II) Determine whether each of the following functions is continuous, and prove your answers.

(a)

$$f(x,y) = \begin{cases} \dfrac{xy}{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

(b)

$$f(x,y) = \begin{cases} \dfrac{x^2 y}{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

Hint. Use polar coordinates.

Problem 2.85. (2022 Fall POSTECH Calculus II) Let

$$f(x,y) = \frac{1}{1-xy}.$$

Find the second-order Taylor approximation to f at $(0,0)$.

Problem 2.86. (2022 Fall POSTECH Calculus II) A line L_1 in $\mathbb{R}^3$ is given by

$$x = t+1, \qquad y = t+2, \qquad z = t+3,$$

and a line L_2 in $\mathbb{R}^3$ is given by

$$x = 1-s, \qquad y = 1+s, \qquad z = 2s.$$

Find a point P on L_1 and a point Q on L_2 such that the square of the distance between P and Q is a local minimum with respect to t and s.

Problem 2.87. (2022 Fall POSTECH Calculus II) Let

$$f(x,y) = x^2 + 3y^2, \qquad g(x,y) = x^4 + 2y^2.$$

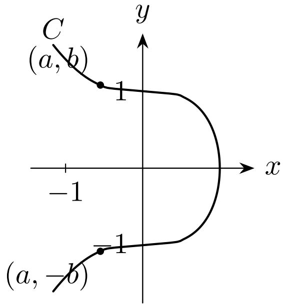
(a) Show that f and g attain their strict local minimum at the origin $(0,0)$.

(b) Show that $\operatorname{Hess} f(0,0)$ is positive definite but $\operatorname{Hess} g(0,0)$ is not positive definite.

Problem 2.88. (2022 Fall POSTECH Calculus II) Let C be the curve in $\mathbb{R}^2$ defined by

$$x^3 + y^2 = 1.$$

The minimum distance from a point on C to the point $(-1, 0)$ occurs at $(a, \pm b) \in C$, where $b > 1$.



Check the conditions for using the Lagrange multiplier method. Find a using the Lagrange multiplier method.

Problem 2.89. (2022 Fall POSTECH Calculus II) Consider the function

$$f(x, y, z) = x^4 + y^4 + z^4 - 4xyz.$$

(a) Find all $P \in \mathbb{R}^3$ such that $\nabla f(P) = \mathbf{0}$.

(b) Show that at each local extremum except the origin, f has a local minimum.

(c) Show that $f(0, 0, 0)$ is not a local extremum.

Problem 2.90. (2022 Fall POSTECH Calculus II) Write a Riemann sum for

$$\iint_{[0,1]\times[0,1]} x^2 y^3 \, \mathrm{d}A$$

using points at the upper-right corner of each $\frac{1}{4}$-square.

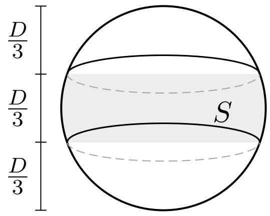
Problem 2.91. (2022 Fall POSTECH Calculus II) Let D be the parallelogram in $\mathbb{R}^2$ bounded by the lines

$$y = -x + 5, \qquad y = -x + 2, \qquad y = 2x - 1, \qquad y = 2x - 4.$$

Compute

$$\iint_D y^2 \, \mathrm{d}A.$$

Problem 2.92. (2022 Fall POSTECH Calculus II) An orange is cut into three slices with equal width.



Assume that the surface of the orange is a sphere with diameter D. Find the area of the peel from the middle slice (the area of S in the figure) by evaluating a surface integral explicitly.

Problem 2.93. (2022 Fall POSTECH Calculus II) Find

$$\oint_C \mathbf{F} \cdot \mathbf{T}\, \mathrm{d}s,$$

where

$$\mathbf{F}(x, y) = \left(-\frac{y}{x^2+y^2}, \frac{x}{x^2+y^2}\right),$$

and C is a loop that contains the origin in its interior and is traversed counterclockwise.

Problem 2.94. (2022 Fall POSTECH Calculus II) Let

$$\mathbf{F}(x, y, z) = (x^2 + y, \sin y, z).$$

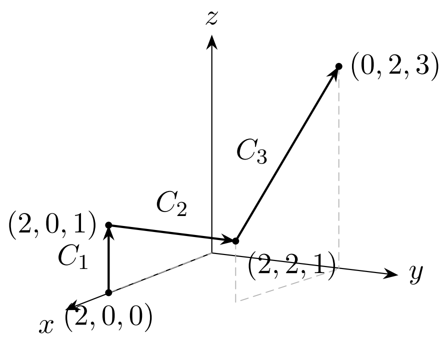
(a) Verify that $\mathbf{F}$ is not conservative.

(b) Verify that

$$\mathbf{G}(x, y, z) = (x^2, \sin y, z)$$

is conservative.

(c) Find the integral of $\mathbf{F}$ along the curve $C = C_1 \cup C_2 \cup C_3$ shown below, using part (b).



Problem 2.95. (2022 Fall POSTECH Calculus II) For each of the following statements, answer True or False. Here C is a smooth curve in $\mathbb{R}^2$, $-C$ is C traversed in the opposite direction, and $\mathbf{T}_C$ and $\mathbf{T}_{-C}$ are the unit tangent vectors to C and $-C$, respectively. The vector field $\mathbf{F}$ is continuously differentiable on $\mathbb{R}^2$.

(1) $\displaystyle\int_C f\ \mathrm{d}s = -\int_{-C} f\ \mathrm{d}s$, where $f\colon \mathbb{R}^2 \to \mathbb{R}$ is continuous.

(2) $\displaystyle\int_C \mathbf{F} \cdot \mathbf{T}_C\ \mathrm{d}s = -\int_{-C} \mathbf{F} \cdot \mathbf{T}_{-C}\ \mathrm{d}s.$

(3) $\int_C \mathbf{F} \cdot \mathbf{N}\ \mathrm{d}s = -\int_{-C} \mathbf{F} \cdot \mathbf{N}\ \mathrm{d}s$ with the same unit normal vector $\mathbf{N}$.

(4) $\mathbf{F}$ is conservative if and only if $\operatorname{curl} \mathbf{F} = 0$.

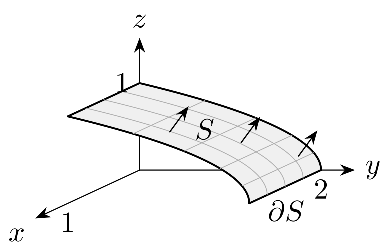
(5)
$$\int_{h(a)}^{h(b)} g(x)\ \mathrm{d}x = \int_a^b g(h(t))|h'(t)|\ \mathrm{d}t,$$
where $g\colon \mathbb{R} \to \mathbb{R}$ is continuous and $h\colon [a,b] \to \mathbb{R}$ is C^1.

Problem 2.96. (2022 Fall POSTECH Calculus II)

(a) State Stokes' theorem, including all conditions.

(b) Let
$$D = \left\{(x,y,z) \in \mathbb{R}^3 : x^2+y^2+z^2 = 5^2,\ y \le -1\right\}.$$
Find the area of D. Verify Stokes' theorem for the vector field
$$\mathbf{F}(x,y,z) = (-z,-4,x).$$

Problem 2.97. (2022 Fall POSTECH Calculus II) Let $\mathbf{F}(x,y,z) = (x,y,z)$, and let $S \subseteq \mathbb{R}^3$ be the surface defined by
$$y + 2z^2 = 2, \qquad 0 \le x \le 1, \qquad 0 \le z \le 1.$$



(a) Find the outward flux of $\mathbf{F}$ across S by evaluating a surface integral.

(b) Find the counterclockwise circulation of $\mathbf{F}$ around ∂S.

Problem 2.98. (2022 Fall POSTECH Calculus II) Let $\mathbf{F} = (xy, z, 2y)$ be a vector field in $\mathbb{R}^3$, and let C be a curve of intersection of the plane
$$x + z = 5$$
and the cylinder
$$x^2 + y^2 = 9.$$
Orient C so that its projection onto the xy-plane is oriented clockwise. Find
$$\int_C \mathbf{F} \cdot \mathbf{T}\ \mathrm{d}s.$$

Problem 2.99. (2022 Fall POSTECH Calculus II) Use each of the following theorems to show that, for a regular set $D \subseteq \mathbb{R}^2$ with boundary oriented counterclockwise,
$$\operatorname{area}(D) = -\int_{\partial D} y\ \mathrm{d}x.$$

(a) Use the Divergence theorem.

(b) Use Green's theorem.

Problem 2.100. (2022 Fall POSTECH Calculus II) Let

$$\mathbf{F}(x, y, z) = (y + z, z + x, x + y)$$

be a vector field in $\mathbb{R}^3$, and let C be a piecewise smooth curve from the origin to any point on the unit sphere

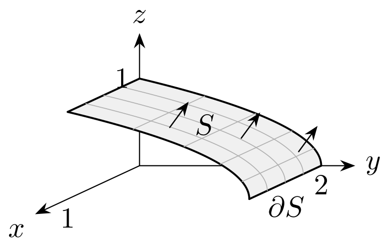
$$x^2 + y^2 + z^2 = 1.$$

Find the maximum of

$$\int_C \mathbf{F} \cdot \mathbf{T}\, \mathrm{d}s.$$

Problem 2.101. (2022 Fall POSTECH Calculus II) Let $\mathbf{F}(x, y, z) = (x, y, z)$, and let $S \subseteq \mathbb{R}^3$ be the surface defined by

$$y + 2z^2 = 2, \qquad 0 \le x \le 1, \qquad 0 \le z \le 1.$$



Using the Divergence theorem, find the outward flux of $\mathbf{F}$ across S.
Hint. Consider a three-dimensional solid whose boundary contains S.

Problem 2.102. (2022 Fall POSTECH Calculus II)

(a) Show that

$$\nabla \cdot (v\nabla u) = v\Delta u + \nabla v \cdot \nabla u$$

for all twice differentiable functions u and v on a regular set $D \subset \mathbb{R}^3$.

(b) Derive the following equality for a twice differentiable function u on D:

$$\iiint_D \left(u\Delta u + \|\nabla u\|^2\right) \mathrm{d}V = \iint_{\partial D} u\nabla u \cdot \mathbf{N}\, \mathrm{d}S,$$

where $\mathbf{N}$ is the unit outward normal vector on the boundary ∂D.

(c) Let u be a twice differentiable function satisfying

$$\Delta u = u \quad \text{in } D, \qquad u = 0 \quad \text{on } \partial D.$$

Show that $u \equiv 0$.

Problem 2.103. (2022 Fall KAIST Calculus II) The K9 Thunder is a Korean 155 mm self-propelled howitzer (cannon). Its maximum range is 18 km. It depends on the type of cannonball fired.

(a) Use the equations for projectile motion studied in class to find the initial speed of the cannonball. Take $g = 9.8\,\mathrm{m/s^2}$.

(b) How many seconds does it take to hit the target at the maximum range?

Problem 2.104. (2022 Fall KAIST Calculus II) Let C be a curve parametrized by

$$\mathbf{r}(t) = (\cos t, \sin t, \sin t \cos t).$$

(a) Find the curvature κ of C at the point

$$\mathbf{r}\left(\frac{\pi}{4}\right) = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{2}\right).$$

(b) Find the torsion τ of C at the point

$$\mathbf{r}\left(\frac{\pi}{3}\right) = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{4}\right).$$

Problem 2.105. (2022 Fall KAIST Calculus II) Let C be the curve

$$\mathbf{r}(t) = (\cos t - 2\sin t + 1, \sin t + 2\cos t, 2t).$$

(a) Find the TNB frame vectors $\mathbf{T}$, $\mathbf{N}$, and $\mathbf{B}$ at $\mathbf{r}(\pi)$.

(b) Find the equation of the osculating plane (the tangent plane orthogonal to $\mathbf{B}$) of C at $\mathbf{r}(\pi)$.

Problem 2.106. (2022 Fall KAIST Calculus II) The gravitational force between the Earth and a satellite is given by

$$\mathbf{F} = -\frac{GmM}{|\mathbf{r}|^2}\frac{\mathbf{r}}{|\mathbf{r}|},$$

where M is the mass of the Earth, m is the mass of the satellite, G is the universal gravitational constant, and $\mathbf{r}$ is the position vector of the satellite when the Earth is at the origin.

(a) Suppose that the satellite orbits the Earth along a circular orbit. Let $r(t) = |\mathbf{r}(t)|$ be the radius and $\theta(t)$ be the angle at time t. Find the formula for the circular motion of the satellite.

(b) Find the escape energy for the satellite, or equivalently the minimum work required to send the satellite to infinity from the Earth's surface. Let R be the radius of the Earth.

Problem 2.107. (2022 Fall KAIST Calculus II) Determine, with justification, whether each of the following limits exists.

(a)

$$\lim_{(x,y)\to(1,1)} \frac{xy^2 - 1}{y - 1}.$$

(b)

$$\lim_{(x,y)\to(1,0)} \frac{xe^y - 1}{xe^y - 1 + y}.$$

Problem 2.108. (2022 Fall KAIST Calculus II) Let

$$f(x,y) = \begin{cases} \dfrac{xy(x^2 - y^2)}{x^2 + y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

(a) Find $\dfrac{\partial f}{\partial y}(x,0)$ for all x and $\dfrac{\partial f}{\partial x}(0,y)$ for all y.

(b) Find

$$\frac{\partial^2 f}{\partial x\,\partial y}(0,0) \qquad \text{and} \qquad \frac{\partial^2 f}{\partial y\,\partial x}(0,0).$$

Problem 2.109. (2022 Fall KAIST Calculus II) Define $f\colon \mathbb{R}^2 \to \mathbb{R}$ by

$$f(0,0) = 0, \qquad f(x,y) = \frac{x(x+y)^2}{y^4+x^2} \quad \text{if } (x,y) \neq (0,0).$$

Let S be the set of all directions (unit vectors) $\mathbf{u} = x\mathbf{i} + y\mathbf{j}$, let $X \subset S$ be the collection of directions for which the directional derivative $D_{\mathbf{u}}f(0,0)$ exists, and define

$$g(x,y) = D_{\mathbf{u}}f(0,0).$$

(a) Find X and $g(x,y)$. Decide whether g is continuous on X.

(b) Find the absolute and local extreme values of g, if they exist.

Problem 2.110. (2022 Fall KAIST Calculus II)

(a) Let

$$f(x,y) = x^2 + 2kxy + 3y^2.$$

For what values of k does f have a local minimum at $(0,0)$? For what values of k does f have a saddle point at $(0,0)$?

(b) Let

$$w = x^2 - y^2 + 4z + t, \qquad x + 2z + t = 25, \qquad x + z + y = 0.$$

Compute $\partial w/\partial x$ when x and y are considered independent variables.

Problem 2.111. (2022 Fall KAIST Calculus II) Let

$$f(x,y) = \sin x \sin y.$$

(a) Use Taylor's formula to find a cubic approximation of f at $(0,0)$, up to third-order terms.

(b) Find an error estimate for the cubic approximation when $|x| < 0.1$ and $|y| < 0.1$, using fourth-order terms.

Problem 2.112. (2022 Fall KAIST Calculus II) Find the minimum distance from the intersection $S = S_1 \cap S_2$ to the origin, where

$$S_1 = \{(x,y,z) : x^2 + 2y^2 - z^2 - 1 = 0\},$$

$$S_2 = \{(x,y,z) : z^2 - 2z + 2 - 2x^2 - y^2 = 0\}.$$

Problem 2.113. (2022 Fall KAIST Calculus II) In this problem, we will prove that the "inverse" Fubini theorem does not work for Riemann integrals and discontinuous functions. Recall that a binary-rational number is a number of the form $n/2^k$, where k and n are integers. Every binary-rational number except 0 can be written in this form with an odd n in a unique way. Let

$$f(x,y) = 1$$

if $x = n/2^k$ and $y = m/2^k$, where n and m are odd integers and k is any integer; note that the same k is used in both expressions. Let $f(x,y) = 0$ otherwise.

(a) For a fixed $y = y_0 \in [0,1]$, what are the values of $x \in [0,1]$ such that $f(x,y_0) = 1$? How many such values are there? Your answer will depend on y_0.
Does the integral

$$\int_0^1 f(x,y_0)\,\mathrm{d}x$$

exist? If so, what is its value?
Does the repeated integral

$$\int_0^1 \int_0^1 f(x,y)\ \mathrm{d}x\ \mathrm{d}y$$

exist?

(b) For a rectangle $R = [a,b] \times [c,d]$, where

$$0 \le a < b \le 1, \qquad 0 \le c < d \le 1,$$

what are the minimum and maximum values of f on R?
Does the double integral

$$\iint_{[0,1]\times[0,1]} f(x,y)\ \mathrm{d}x\ \mathrm{d}y$$

exist?

Problem 2.114. (2022 Fall KAIST Calculus II) Let

$$D = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 \le 2x \text{ and } x^2 + y^2 \le 2\sqrt{3}\,y\}$$

be the region in the xy-plane.

(a) Split

$$\iint_D f(x,y)\ \mathrm{d}A$$

into a sum of two iterated integrals in polar form.

(b) Find the area of D.

Problem 2.115. (2022 Fall KAIST Calculus II) Let R be a region in the first octant $(x, y, z > 0)$ in $\mathbb{R}^3$ satisfying

$$x + y^2 + 4z^2 \le 4.$$

(a) Describe the triple integral

$$\iiint_R f(x,y,z)\ \mathrm{d}V$$

as an iterated integral (i) in the order $\mathrm{d}x\ \mathrm{d}y\ \mathrm{d}z$ and (ii) in the order $\mathrm{d}y\ \mathrm{d}z\ \mathrm{d}x$.

(b) Evaluate

$$\iiint_R yz\ \mathrm{d}V.$$

Problem 2.116. (2022 Fall KAIST Calculus II) Let R be a region in the first quadrant of the xy-plane bounded by

$$xy = 4, \qquad xy = 16, \qquad y = x, \qquad y = 9x.$$

(a) Use the transformation

$$x = \frac{u}{v}, \qquad y = uv, \qquad u > 0, \qquad v > 0,$$

and find a region G in the uv-plane and a function $f(u,v)$ such that

$$\iint_R \left(\sqrt{\frac{y}{x}} + \sqrt{xy}\right) \mathrm{d}x\ \mathrm{d}y = \iint_G f(u,v)\ \mathrm{d}u\ \mathrm{d}v.$$

(b) Evaluate the above integral.

Problem 2.117. (2022 Fall KAIST Calculus II) Let

$$\mathbf{\Phi}(u, v) = u \sin v\, \mathbf{i} - v\, \mathbf{j} + u \cos v\, \mathbf{k}, \qquad 0 \leq u \leq 1, \qquad 0 \leq v \leq \pi.$$

(a) Describe the smooth component curves of the boundary curve of the surface defined by $\mathbf{\Phi}$. Identify the one, say C, with maximum arc length.

(b) Let

$$f(x, y, z) = z^4 e^x - 3(e^x + 3y)xz^2.$$

Find the point on C that divides C into two curves C_1 and C_2 of equal arc length. Evaluate the line integrals of f along the curves C_1 and C_2.

Problem 2.118. (2022 Fall KAIST Calculus II) Let $\mathbf{F} = M\mathbf{i} + N\mathbf{j}$, and let C be a simple, closed curve enclosing a region D in the plane.

(a) State the two versions of Green's Theorem and derive one version from the other.

(b) Let

$$M = \frac{x^4}{4}, \qquad N = x^3 y, \qquad C = \left\{(x, y) \in \mathbb{R}^2 : \frac{x^4}{a^4} + \frac{y^2}{b^2} = 1\right\}.$$

Find the outward flux

$$\int_C \mathbf{F} \cdot \mathbf{n}\ \mathrm{d}s = \int_C M\ \mathrm{d}y - N\ \mathrm{d}x.$$

Problem 2.119. (2022 Fall KAIST Calculus II) Let

$$\mathbf{F} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$

and let S be the ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1, \qquad a, b, c > 0.$$

(a) Use spherical coordinates (ϕ, θ) and find a parametrization of S in the form

$$\mathbf{r}(\phi, \theta) = (x(\phi, \theta), y(\phi, \theta), z(\phi, \theta)),$$

and compute $\mathbf{r}_\phi \times \mathbf{r}_\theta$.

(b) Compute $\mathbf{n}\ \mathrm{d}S$ and the outward flux

$$\iint_S \mathbf{F} \cdot \mathbf{n}\ \mathrm{d}S.$$

Problem 2.120. (2022 Fall KAIST Calculus II) Let $\mathbf{F} = M\mathbf{i} + N\mathbf{j} + P\mathbf{k}$ be continuously differentiable, let

$$S = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\}$$

be the unit sphere, let

$$S_1 = \{(x, y, z) \in S : z \geq 0\}, \qquad S_2 = \{(x, y, z) \in S : z \leq 0\}$$

be the upper and lower hemispheres, and let

$$B = \{(x, y, z) : x^2 + y^2 + z^2 \leq 1\}$$

be the unit ball.

(a) Use the fact that $S = S_1 \cup S_2$ and Stokes' Theorem to show that

$$\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, \mathrm{d}S = 0.$$

(b) Use the Divergence Theorem to show that

$$\iiint_B \nabla \cdot (\nabla \times \mathbf{F}) \, \mathrm{d}V = 0.$$

Problem 2.121. (2022 Fall KAIST Calculus II) Let $(x, y) \in \mathbb{R}^2$ be fixed, let $\varepsilon > 0$ be small, let

$$A_\varepsilon = [x, x + \varepsilon] \times [y, y + \varepsilon]$$

be a small square cell, and let $C_\varepsilon = \partial A_\varepsilon$ be its boundary. Let $\mathbf{F} = M\mathbf{i} + N\mathbf{j}$ be twice differentiable and suppose that

$$|M_{xx}|, |M_{xy}|, |M_{yy}|, |N_{xx}|, |N_{xy}|, |N_{yy}| \leq K_0,$$

i.e., all second-order partial derivatives of M and N are bounded by K_0. Note that the area of A_ε is $\Delta(A_\varepsilon) = \varepsilon^2$. Use Taylor's formula and show that

$$\left| \int_{C_\varepsilon} \mathbf{F} \cdot \mathbf{n} \, \mathrm{d}s - (M_x(x, y) + N_y(x, y))\varepsilon^2 \right| \leq 4K_0\varepsilon^3.$$

Problem 2.122. (2022 Fall KAIST Calculus II) Let D be a right cylinder, perpendicular to the xy-plane, whose base is the region between the circles

$$r = \cos\theta \qquad \text{and} \qquad r = 2\cos\theta,$$

and whose top lies in the plane

$$z = 4 - x - y.$$

Find the volume of D.

Problem 2.123. (2023 Fall POSTECH Calculus II) Find an equation for the plane tangent to

$$x - 3y^2 + z^2 = 7$$

at $(1, 1, 3)$.

Problem 2.124. (2023 Fall POSTECH Calculus II) Let L be the line containing $(1, 2, 1, 1)$ and $(1, 0, 2, 2)$ in $\mathbb{R}^4$. Let P be the hyperplane defined by

$$x + y + z + w = 1$$

in $\mathbb{R}^4$.

(a) Do L and P intersect each other?

(b) If the answer is yes, find the intersection point. Otherwise, find the distance between L and P.

Problem 2.125. (2023 Fall POSTECH Calculus II) Let $f\colon \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$f(x, y) = |x^3 + y^3|.$$

(a) Find $\nabla f(0, 0)$.

(b) Is f differentiable at $(0, 0)$? Why or why not? Justify your answer.

Problem 2.126. (2023 Fall POSTECH Calculus II) Let $u\colon \mathbb{R}^2 \to \mathbb{R}$ be a C^2 function, and define

$$v(r,\theta) := u(r\cos\theta, r\sin\theta).$$

Show that

$$u_{xx} + u_{yy} = v_{rr} + \frac{1}{r}v_r + \frac{1}{r^2}v_{\theta\theta}.$$

Problem 2.127. (2023 Fall POSTECH Calculus II) Suppose the electric potential at (x,y) is given by

$$\ln\left(\sqrt{x^2+y^2}\right).$$

(a) Find the rate of change of the potential at $(3,4)$ toward the origin.

(b) Find the rate of change of the potential at $(3,4)$ in a direction perpendicular to the direction given above. You may choose either of the two possible directions.

Problem 2.128. (2023 Fall POSTECH Calculus II) Suppose there is a function $\mathbf{F}\colon \mathbb{R}^4 \to \mathbb{R}^2$ with component functions $\mathbf{F} = (f_1, f_2)$ defined by

$$\begin{aligned} f_1(x,y,z,w) &= e^{x+2y-z+w} - (2x^2 - y^2 + z^2 - 3w^2), \\ f_2(x,y,z,w) &= \ln(1 + x^2 + y^2 + z^2 + w^2) + (3x + 4y - 5z + w). \end{aligned}$$

Let $P = (0,0,0,0) \in \mathbb{R}^4$.

(a) Show that there is a differentiable function $\mathbf{G}\colon \mathbb{R}^2 \to \mathbb{R}^2$ defined on a small open set U around $(0,0)$ satisfying $\mathbf{G}(0,0) = (0,0)$ and

$$\mathbf{F}(P) = \mathbf{F}(x,y,\mathbf{G}(x,y))$$

for $(x,y) \in U$.

(b) Using the linear approximation of $\mathbf{G}$, find an approximate solution (z,w) of the nonlinear system

$$\begin{aligned} 1 &= e^{0.1+2(0.1)-z+w} - (2(0.1)^2 - (0.1)^2 + z^2 - 3w^2), \\ 0 &= \ln(1 + (0.1)^2 + (0.1)^2 + z^2 + w^2) + (3(0.1) + 4(0.1) - 5z + w). \end{aligned}$$

Problem 2.129. (2023 Fall POSTECH Calculus II) Find the shortest distance from $(0,0,3)$ to the surface

$$z = x^2 - y^2.$$

Problem 2.130. (2023 Fall POSTECH Calculus II) Find the local maximum or minimum value of the function

$$x^3 + y^3 + z^3 - 3xy - 3yz - 3zx$$

defined on the domain

$$\{(x,y,z) : x > 0,\ y > 0,\ z > 0\}.$$

Problem 2.131. (2023 Fall POSTECH Calculus II) Show that the function

$$f(x,y) = x + y$$

subject to the constraint

$$x^3 + x + y^3 + y = 1$$

has at most one local extremum.

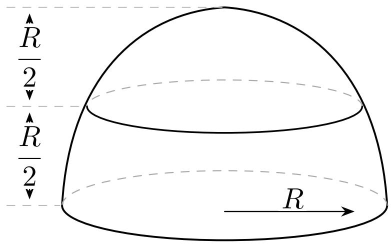
Problem 2.132. (2023 Fall POSTECH Calculus II) Let $D \subseteq \mathbb{R}^2$ be the domain given by

$$\frac{(x-1)^2}{4} + \frac{(y-3)^2}{9} \leq 1.$$

Find

$$\iint_D (x+y)\,\mathrm{d}A.$$

Problem 2.133. (2023 Fall POSTECH Calculus II) A solid hemisphere of radius R is split into two pieces with equal thickness. Use spherical coordinates to find the ratio of the volumes of the two pieces. Your answer should be a number between 0 and 1.



Problem 2.134. (2023 Fall POSTECH Calculus II) Let $\mathbf{F}$ be a vector field on $\mathbb{R}^2 \setminus \{(0,0)\}$ defined by

$$\mathbf{F}(x,y) = \left(\frac{x}{x^2+y^2}, \frac{y}{x^2+y^2}\right).$$

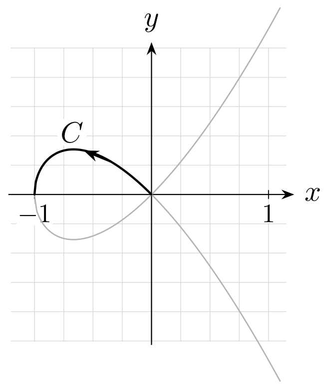
(a) Show that $\operatorname{curl}\mathbf{F} = 0$.

(b) Is $\mathbf{F}$ conservative? If so, find its potential. If not, explain why.

Problem 2.135. **(2023 Fall POSTECH Calculus II)** Let $C \subseteq \mathbb{R}^2$ be the curve defined by

$$y^2 = x^3 + x^2$$

in the second quadrant ($-1 \le x \le 0,\ y \ge 0$), oriented "to the left."



Find the work done by the force

$$\mathbf{F} = (2xy, -x^3)$$

along C.

Problem 2.136. **(2023 Fall POSTECH Calculus II)** Let S be the surface formed by the part of the paraboloid

$$z = 1 - x^2 - y^2$$

lying above the xy-plane, and let

$$\mathbf{F} = (x, y, 2(1-z)).$$

Calculate the upward flux of $\mathbf{F}$ across S in each of the following ways:

(a) By direct calculation of

$$\iint_S \mathbf{F} \cdot \mathbf{N}\, \mathrm{d}S.$$

(b) Using the Divergence theorem and computing the flux across a simpler surface.

Problem 2.137. **(2023 Fall POSTECH Calculus II)** Let

$$\mathbf{F}(x, y, z) = (yz, -xz, 1).$$

Let S be the portion of the surface of the paraboloid

$$z = 4 - x^2 - y^2$$

which lies above the first octant, and let C be the closed curve

$$C = C_1 + C_2 + C_3,$$

where C_1, C_2, C_3 are the three curves formed by intersecting S with the xy-, yz-, and xz-planes, respectively, so that C is the boundary of S. Orient C so that it is traversed counterclockwise when seen from above in the first octant. Evaluate

$$\int_C \mathbf{F} \cdot \mathbf{T}\, \mathrm{d}s.$$

Problem 2.138. (2023 Fall POSTECH Calculus II) Let $\mathbf{F}$ be the vector field defined by

$$\mathbf{F}(x,y,z) = \left(\frac{z}{x^2+z^2}, y, -\frac{x}{x^2+z^2} \right).$$

Let C be the intersection of

$$x^2+z^2=4 \qquad \text{and} \qquad x+y+z=0$$

in $\mathbb{R}^3$. Find the circulation of $\mathbf{F}$ over C, oriented from $(2,-2,0)$ to $(0,-2,2)$, counterclockwise when viewed from the positive y-axis.

Problem 2.139. (2023 Fall POSTECH Calculus II) Two point charges are placed at

$$P_1 = (1,0,0) \qquad \text{and} \qquad P_2 = (-1,0,0),$$

with electric charges q_1 and q_2, respectively. The electric field $\mathbf{F}$ on $\mathbb{R}^3 \setminus \{P_1, P_2\}$ exerted by these two particles is given by

$$\mathbf{F}(x,y,z) = \frac{q_1}{4\pi\varepsilon_0} \frac{(x-1,y,z)}{\left((x-1)^2+y^2+z^2\right)^{3/2}} + \frac{q_2}{4\pi\varepsilon_0} \frac{(x+1,y,z)}{\left((x+1)^2+y^2+z^2\right)^{3/2}},$$

where ε_0 is the electric constant. If S is the ellipsoid

$$\frac{x^2}{2} + y^2 + z^2 = 1,$$

write the outward flux

$$\iint_S \mathbf{F} \cdot \mathbf{N} \, \mathrm{d}S$$

in terms of q_1 and q_2. Justify your answer.

Problem 2.140. (2024 Fall POSTECH Calculus II) Find parametric equations for the line of intersection of the planes

$$3x+2y-z=5 \qquad \text{and} \qquad x+y+z=2.$$

Also find the distance from the point $S(1,2,2)$ to this line.

Problem 2.141. (2024 Fall POSTECH Calculus II) Find the directional derivative of

$$f(x,y,z) = xy^2 + yz^3$$

at $P(2,-1,1)$ in the direction of $\overrightarrow{PQ}$, where $Q=(3,1,3)$.

Problem 2.142. (2024 Fall POSTECH Calculus II) Explicitly compute

$$\nabla \cdot (\nabla f) \qquad \text{and} \qquad \nabla \times (\nabla f)$$

for

$$f(x,y,z) = x^3+y^3+z^3-3xyz.$$

Problem 2.143. (2024 Fall POSTECH Calculus II) Let $D, E \subseteq \mathbb{R}^n$. Prove the following:

(a) If D and E are open, then $D \cap E$ is open.

(b) If $D \subseteq E$, then $\operatorname{int} D \subseteq \operatorname{int} E$, where $\operatorname{int} D$ denotes the interior of D.

Problem 2.144. (2024 Fall POSTECH Calculus II) Let $f\colon \mathbb{R} \to \mathbb{R}$ be continuous. Show that for any $i = 1, \ldots, n$, the function

$$F_i\colon \mathbb{R}^n \to \mathbb{R}, \qquad (x_1, \ldots, x_n) \mapsto f(x_i),$$

is continuous.

Problem 2.145. (2024 Fall POSTECH Calculus II)

(a) Show that

$$g(x,y) = \begin{cases} \dfrac{x^2 - y^2}{x^2 + y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0), \end{cases}$$

is not continuous at $(0,0)$.

(b) Show that

$$f(x,y) = \begin{cases} \dfrac{xy(x^2 - y^2)}{x^2 + y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0), \end{cases}$$

is differentiable at $(0,0)$.

Problem 2.146. (2024 Fall POSTECH Calculus II) Let $x, y\colon \mathbb{R}^2 \to \mathbb{R}$ be functions defined by

$$x(u,v) = u^3 - 3uv^2,$$
$$y(u,v) = 3u^2 v - v^3.$$

Also define

$$f(x,y) = e^{x^2 - y^2} \cos(2xy).$$

(a) Show that

$$x_u = y_v, \qquad x_v = -y_u, \qquad f_{xx} + f_{yy} = 0.$$

(b) Let

$$g(u,v) := f\big(x(u,v), y(u,v)\big).$$

Compute $g_{uu} + g_{vv}$.

Problem 2.147. (2024 Fall POSTECH Calculus II) Find all the local maxima, local minima, and saddle points of

$$f(x,y) = e^x(2x^2 - y^2).$$

Problem 2.148. (2024 Fall POSTECH Calculus II) Let

$$T(x,y,z) = 2x^2 + 4yz - 8z + 300$$

be the temperature at the point (x,y,z). Find the hottest and coldest points on the ellipsoid

$$x^2 + y^2 + 2z^2 = 4$$

and their temperatures. Use the Lagrange multiplier method.

Problem 2.149. (2024 Fall POSTECH Calculus II) Find the order 2 Taylor approximation of

$$f(x,y,z) = \cos(x+y+z)$$

at $P = (0,0,0)$, and use it to approximate

$$f(0.1,0.1,0.1) = \cos(0.3).$$

Also find an error bound for the approximation.

Problem 2.150. (2024 Fall POSTECH Calculus II) Let $\mathbf{F}\colon \mathbb{R}^3 \to \mathbb{R}^2$, $\mathbf{F} = (f_1, f_2)$, be defined by

$$\begin{aligned} f_1(x,y,z) &= x^2 + y^2 - z^2, \\ f_2(x,y,z) &= \frac{x^2}{5} + \frac{y^2}{2} + \frac{z^2}{2}. \end{aligned}$$

Let

$$P = \left(1, \frac{2}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right).$$

Then the level set

$$C = \{(x,y,z) \in \mathbb{R}^3 \mid \mathbf{F}(x,y,z) = \mathbf{F}(P)\}$$

defines a curve in $\mathbb{R}^3$.

(a) Show that there is a continuously differentiable function $\mathbf{G}\colon \mathbb{R} \to \mathbb{R}^2$, $\mathbf{G} = (g_1, g_2)$, such that every point in C sufficiently close to P can be expressed as

$$(x,y,z) = (\mathbf{G}(z), z).$$

(b) Find any unit tangent vector tangent to C at P.

3 (MATH311) Analysis I

4 (MATH312) Analysis II